GREEN BIOTECHNOLOGY & PACKAGING INNOVATION

Plant-Based Spoilage Indicator Sticker

A 100% natural, color-changing sensor that shows if food or medicine has gone bad without opening the package.

1. THE REAL PROBLEM

Most of us rely on the printed "Best Before" date on a carton or bottle, but those calendar dates are just rough estimates. Because of that, families, supermarkets, and clinics toss out millions of pounds of perfectly good perishables every week.

Perishable Foods & Cold Chains

- **Invisible Toxins:** If food is left warm during transit, dangerous bacteria generate toxins long before odors or visible mold appear.
- **Costly Guesswork:** Safe food is discarded simply because a date passed, while spoiled food is eaten because the date says it is fine.

Sensitive Medicine & Vaccines

- **Quiet Breakdown:** Insulin, vaccines, liquid antibiotics, and eye drops lose potency when exposed to heat or humidity fluctuations.
- **No Visible Change:** Degraded medicine rarely changes appearance, putting patients at risk of untreated illness.

2. OUR SOLUTION: BIO-CHROMA STICKER

Core Idea: We developed a non-toxic, compostable smart sticker made entirely from food-grade plant extracts and biopolymer films. Placed inside packaging headspace, it reacts to spoilage gases and permanently changes color—acting like a direct traffic light so anyone can check freshness at a glance.

3. HOW IT WORKS

- **Gas & Biomarker Release:** As perishables spoil, microbial metabolic reactions produce volatile organic vapors like ammonia (NH_3), dimethylamine, Total Volatile Basic Nitrogen ($TVB\text{-}N$), and organic acids.
- **Breathable Shield:** A hydrophobic, microporous base membrane lets headspace vapors drift through to the dye while blocking liquid splashes and condensation from causing false alarms.
- **Natural Color Shift:** Plant anthocyanins shift molecular structure across pH levels—moving from red flavylium cations ($pH < 3$) to purple quinoidal bases ($pH\ 6–7$), and finally to greenish-yellow chalcones ($pH > 8$).

4. VISUAL STATUS GUIDE

STATUS	INDICATOR COLOR	BIOCHEMICAL CONDITION	RECOMMENDED ACTION
FRESH / SAFE	Rich Red / Pink	Normal acidic/neutral state; low volatiles	Safe to use or consume
CAUTION	Purple / Violet	Early microbial breakdown; mild pH rise	Use immediately; inspect storage
SPOILED / DEGRADED	Greenish-Blue / Yellow	High amine accumulation; alkaline shift	Do not consume; discard safely

5. SUSTAINABLE MATERIALS BREAKDOWN

COMPONENT	NATURAL MATERIAL	KEY PURPOSE
Color Sensors	Anthocyanins (red cabbage, blueberry, butterfly pea) & Curcumin	Natural plant pigments that change color in response to pH and volatile gases.
Holding Matrix		

COMPONENT	NATURAL MATERIAL	KEY PURPOSE
	Chitosan, Agar-agar, Sodium Alginate, or Cornstarch + Glycerol	Holds the plant dyes firmly, prevents fading, and keeps the film flexible.
Protective Layer	Cellulose acetate with hydrophobic beeswax coating	Allows target gases to enter while preventing moisture and liquid contact.
Cover & Guide	Polylactic Acid (PLA) film + Soy-ink reference ring & Starch adhesive	Clear window for consumer inspection, safe adhesion, and instant color comparison.

6. WHY IT MATTERS

- **Zero Guesswork:** Protects consumers from food poisoning and ensures patients never receive inactive medication.
- **Completely Eco-Friendly:** Zero electronic waste, 100% compostable, and safe to use right next to food.
- **Fraction-of-a-Cent Cost:** Highly scalable and manufacturable at tiny unit costs for mass packaging adoption.